

HCSM-54

Vacuum Stability and the Stability Coefficient

The AGM Closed Form, the Self-Dual Torus,
and the Crossover Zero Moments

Hodge Complex Standard Model

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Abstract

We establish the vacuum stability sector of the Hodge Complex Standard Model. The stability coefficient

$$\gamma = c_2/I(2) = 0.70283946799007245929257819650854315216110316215188\dots$$

controls the width of the vacuum entropy well $S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4)$ around the equilibrium dimension $D = 4$, and the strict local maximum property $S''(0)/S_0 = -2\gamma < 0$ is the vacuum stability theorem. We prove four results. (i) The strip integral at $\alpha = 2$ has the exact closed form

$$I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2}),$$

verified at 100-digit precision. (ii) The AGM constant is a substrate level invariant of the self-dual square torus: the 8×8 lattice is square, its modular parameter is $\tau = i$, the Dedekind eta function at the self dual point is $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$ by the Chowla–Selberg formula, and

$$\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2).$$

This is a derivation of the transcendental content of $I(2)$ from the modular structure of the substrate. (iii) The second derivative $c_2 = 2I''(2)$ has an exact closed form in π , $\Gamma(1/4)$, and Catalan's constant G . (iv) The Brillouin-zone moments of the three crossover-zero integrands on the 8×8 torus are exact integers 4, 12, 28, following the pattern $d(2^n - 1)$ for $n = 1, 2, 3$ with $d = 4$, and the mirror ratio $1/3 = 1/(d - 1)$ is the generation count denominator.

A negative result is also established: the discrete analogue of $I(2)$ is not a Brillouin zone moment of a crossover vanishing integrand. The closest candidate, the spectral sum $\langle \sqrt{\lambda}/(1 + \lambda) \rangle_{\text{BZ}} = 0.40083787\dots$, matches $I(2) = 0.40092988\dots$ to 0.023%; the residual is not a framework constant. The HCSM-native derivation of $I(2)$ therefore proceeds through the modular structure of the self-dual square torus (result ii).

Derivation status summary:

- **DERIVED (substrate-level):** the self-dual modular structure of the 8×8 torus ($\tau = i$); the Chowla–Selberg value $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$; the AGM identity $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$; the substrate level form $I(2) = 1 - 1/(2\sqrt{2}\eta(i)^2)$; the crossover zero moments 4, 12, 28; the mirror ratio $1/3 = 1/(d - 1)$.

- **DERIVED (analytic):** the AGM closed form $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ from the Beta function; the closed form of $c_2 = 2I''(2)$ in π , $\Gamma(1/4)$, G ; the closed form of $\gamma = c_2/I(2)$.
- **DISCLOSED IDENTIFICATION:** the identification $\gamma = c_2/I(2)$ of the stability coefficient with the ratio of the second derivative $c_2 = 2I''(2)$ to the value $I(2)$ of the strip integral at $\alpha = 2$.
- **NEGATIVE RESULT:** the discrete analogue of $I(2)$ is not a Brillouin-zone moment of a crossover-vanishing integrand.
- **OPEN:** the physical mechanism by which c_2 enters at the second order of the entropy well; the finite- L corrections to the crossover zero moments; the connection to the sigma-line structure of HCSM-24.

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1 Introduction

1.1 The vacuum stability problem

The Hodge Complex Standard Model (HCSM) derives Standard Model structure from the spacetime dimension $d = 4$ through the discrete substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ (HCSM-00, HCSM-01, HCSM-02). The vacuum stability sector supplies the framework’s stability mechanism: it shows that $D = 4$ is a stable fixed point of the vacuum entropy, and it computes the exact stability coefficient γ that controls the width of the entropy well.

The stability coefficient enters the framework’s downstream predictions in three places. First, the EFT vacuum equation of state is $w = -1 - (1 - \gamma)^2 / (2\pi)$, giving $w = -1.0140540788576648688 \dots$, consistent with DESI DR2 at $\Delta\chi^2 = 0.22$. Second, the bulk viscosity is $\zeta H / \rho = (1 - \gamma)^2 / (6\pi)$. Third, the time quantum in the Motion and Time Theorems is $T = 1 / (2\gamma)$ (HCSM-20, HCSM-21).

The strip integral $I(\alpha)$ is a continuum object; the framework’s strength is that it derives Standard Model structure from a discrete substrate, so a continuum object at the root of the entropy well construction is a genuine open item.

The present paper addresses this. It establishes four structural facts:

1. **Closed form of $I(2)$ and γ** (Theorems 3.1, 3.3). The strip integral at $\alpha = 2$ is the halved arithmetic-geometric mean of 1 and $\sqrt{2}$, subtracted from 1. The second derivative c_2 has a closed form in π , $\Gamma(1/4)$, and Catalan’s G . The stability coefficient is their exact ratio.
2. **Substrate-level derivation of $\text{AGM}(1, \sqrt{2})$** (Theorem 3.4). The 8×8 torus is square, its modular parameter is $\tau = i$, the Dedekind eta function at the self dual point is $\eta(i) = \Gamma(1/4) / (2\pi^{3/4})$, and $\text{AGM}(1, \sqrt{2}) = 1 / (\sqrt{2} \eta(i)^2)$. The transcendental content of $I(2)$ is derived from the modular structure of the substrate.
3. **Crossover-zero moments** (Theorem 4.1, Corollary 4.3). The Brillouin zone moments of the three crossover-zero integrands on the torus are exact framework integers, and their mirror ratio is the generation count denominator.
4. **Negative result** (Section 5). The discrete analogue of $I(2)$ is not a BZ moment of a crossover-vanishing integrand. The closest candidate, the S6 spectral sum, is 0.023% low; the residual is not a framework constant.

1.2 Summary of results

1. (Theorem 3.1). The strip integral at $\alpha = 2$ satisfies

$$I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2}),$$

verified at 100-digit precision.

2. (Theorem 3.2). The second derivative $c_2 = 2I''(2)$ has the closed form

$$c_2 = 4 + B(-1/4, 1/2) \left[\frac{\pi+4}{8} + \frac{(\pi+4)^2}{128} + \frac{1}{4} + \frac{1-G}{8} \right],$$

where $B(-1/4, 1/2) = -4\pi^{3/2}\sqrt{2}/\Gamma(1/4)^2$ and G is Catalan's constant.

3. (Theorem 3.3). The stability coefficient is

$$\gamma = c_2/I(2) = 0.70283946799007245929257819650854315216110316215188\dots$$

4. (Theorem 3.4). The AGM constant is a substrate-level invariant of the self-dual square torus:

$$\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2),$$

with $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$. Corollary 3.5 gives the substrate-level form $I(2) = 1 - 1/(2\sqrt{2}\eta(i)^2)$.

5. (Theorem 4.1). The Brillouin-zone moments of the three crossover-zero integrands satisfy

$$\langle(\lambda - 4)^2\rangle_{\text{BZ}} = 4, \quad \langle\lambda(8 - \lambda)\rangle_{\text{BZ}} = 12, \quad \langle(\lambda - 4)^2\lambda(8 - \lambda)\rangle_{\text{BZ}} = 28.$$

6. (Corollary 4.3). The mirror ratio $\langle(\lambda - 4)^2\rangle/\langle\lambda(8 - \lambda)\rangle = 1/3 = 1/(d - 1)$.

7. (Theorem 5.1). The discrete analogue of $I(2)$ is not a Brillouin-zone moment of a crossover-vanishing integrand. The closest candidate $\langle\sqrt{\lambda}/(1 + \lambda)\rangle_{\text{BZ}} = 0.40083787\dots$ is 0.023% below $I(2)$.

1.3 Derivation status

DERIVED (substrate-level). The self-dual modular structure of the 8×8 torus ($\tau = i$); the Chowla–Selberg value $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$; the AGM identity $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$; the substrate-level form $I(2) = 1 - 1/(2\sqrt{2}\eta(i)^2)$; the crossover-zero moments 4, 12, 28; the mirror ratio $1/3 = 1/(d - 1)$.

DERIVED (analytic). The AGM closed form $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ from the Beta function; the closed form of $c_2 = 2I''(2)$ in π , $\Gamma(1/4)$, and G ; the closed form of $\gamma = c_2/I(2)$.

DISCLOSED IDENTIFICATION. The identification $\gamma = c_2/I(2)$ of the stability coefficient with the ratio of the second derivative $c_2 = 2I''(2)$ to the value $I(2)$ of the strip integral at $\alpha = 2$.

NEGATIVE RESULT. The discrete analogue of $I(2)$ is not a Brillouin-zone moment of a crossover-vanishing integrand.

OPEN. The physical mechanism by which c_2 (rather than another derivative) enters at the second order of the entropy well; the finite- L corrections to the crossover-zero moments; the connection between the crossover-zero integrands and the sigma-line structure of HCSM-24.

1.4 Organization

Section 2 sets up the strip integral and the stability coefficient. Section 3 proves the AGM closed form and gives the substrate-level derivation of $\text{AGM}(1, \sqrt{2})$ from the self-dual square torus. Section 4 proves the crossover-zero moments. Section 5 establishes the negative result on the discrete analogue. Section 6 states the vacuum stability theorem. Section 7 catalogues the derivation status. Section 8 lists falsifiable predictions. Appendix A provides reproducible Python verification. Appendix B gives structured bibliographic metadata.

2 Setup: the strip integral and the stability coefficient

2.1 The dimensional deviation

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The dimension of the vacuum manifold is $D = 4$. The dimensionless deviation from the equilibrium dimension is

$$\epsilon = D - 4. \quad (1)$$

The vacuum entropy well is

$$S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4), \quad (2)$$

with γ the stability coefficient. The sign of the second derivative at $\epsilon = 0$ is -2γ , so $D = 4$ is a strict local maximum of the vacuum entropy.

2.2 The strip integral

The strip integral at parameter α is

$$I(\alpha) = \int_1^\infty \left[\frac{u^{2\alpha-2}}{\sqrt{u^{2\alpha}-1}} - u^{\alpha-2} \right] du. \quad (3)$$

It has the closed form

$$I(\alpha) = \frac{1}{2\alpha} \left[B(a, 1/2) - \frac{1}{a} \right], \quad a = -\frac{1}{2} + \frac{1}{2\alpha}. \quad (4)$$

At $\alpha = 2$ (the physical value $\alpha = D - 2 = 2$), this gives $a = -1/4$ and

$$I(2) = \frac{1}{4} B(-1/4, 1/2) + 1. \quad (5)$$

2.3 The stability coefficient

The stability coefficient is

$$\gamma = \frac{c_2}{I(2)}, \quad c_2 = 2I''(2), \quad (6)$$

where the prime denotes differentiation with respect to α . The numerical value is

$$\gamma = 0.70283946799007245929257819650854315216110316215188 \dots$$

The identification $\gamma = c_2/I(2)$ is the framework's rigidity structure. It is disclosed in the present paper, and the AGM/closed-form work below shows that both $I(2)$ and c_2 are specific transcendental constants rather than opaque integrals.

3 The AGM Closed Form

3.1 The strip integral at $\alpha = 2$

Theorem 3.1 (AGM identity, analytic form). *The strip integral at $\alpha = 2$ satisfies*

$$I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2}), \quad (7)$$

where AGM denotes the arithmetic-geometric mean. Equivalently,

$$I(2) = 1 - \frac{\sqrt{2} \pi^{3/2}}{\Gamma(1/4)^2} = 1 - \frac{\pi}{2\sqrt{2} K(1/\sqrt{2})}, \quad (8)$$

with K the complete elliptic integral of the first kind.

Sketch. From Eq. (5), $I(2) = 1 + B(-1/4, 1/2)/4$. The Beta function is evaluated by reflection:

$$B(-1/4, 1/2) = \frac{\Gamma(-1/4)\Gamma(1/2)}{\Gamma(1/4)} = -\frac{4\Gamma(3/4)\sqrt{\pi}}{\Gamma(1/4)}.$$

Using $\Gamma(1/4)\Gamma(3/4) = \pi/\sin(\pi/4) = \pi\sqrt{2}$, this gives

$$B(-1/4, 1/2) = -\frac{4\sqrt{2} \pi^{3/2}}{\Gamma(1/4)^2}.$$

The classical identity $\text{AGM}(1, \sqrt{2}) = \sqrt{2} \pi^{3/2}/\Gamma(1/4)^2$ then gives the result. The remaining equivalences follow from $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$. $\square$

3.2 The second derivative

Theorem 3.2 (Closed form of c_2). *The second derivative $c_2 = 2I''(2)$ has the closed form*

$$c_2 = 4 + B(-1/4, 1/2) \left[\frac{\pi + 4}{8} + \frac{(\pi + 4)^2}{128} + \frac{1}{4} + \frac{1 - G}{8} \right], \quad (9)$$

with $B(-1/4, 1/2)$ as above and G Catalan's constant.

Sketch. Differentiate Eq. (4) twice with respect to α . The result is expressible in terms of $B(a, 1/2)$ at $a = -1/4$, its α -derivatives, and the polygamma combinations $\psi_1(1/4)$ and $\psi_1(-1/4)$. Using $\psi_1(1/4) = \pi^2 + 8G$ and the reflection $\psi_1(-1/4) = \pi^2 + 8G + 16(1 - G)$ yields the stated form. Verified at machine precision in Appendix A. $\square$

3.3 The stability coefficient

Theorem 3.3 (Closed form of γ). *The stability coefficient is*

$$\gamma = \frac{c_2}{I(2)} = 0.70283946799007245929257819650854315216110316215188 \dots \quad (10)$$

i.e., a specific transcendental constant expressible in π , $\Gamma(1/4)$, and G .

Proof. Direct from Theorems 3.1 and 3.2. $\square$

3.4 The self-dual torus and the Dedekind eta function

Theorem 3.4 (Substrate-level derivation of $\text{AGM}(1, \sqrt{2})$). *The arithmetic-geometric mean $\text{AGM}(1, \sqrt{2})$ that appears in Theorem 3.1 is a substrate-level invariant of the 8×8 torus. Specifically:*

1. *The 8×8 lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ is square: the lattice spacing is equal in the two directions. Its complex structure is therefore the modular parameter*

$$\tau = i. \quad (11)$$

The point $\tau = i$ is the unique self-dual point of the modular group $\text{PSL}(2, \mathbb{Z})$: it is fixed by the S -transformation $\tau \mapsto -1/\tau$.

2. *The Dedekind eta function at the self-dual point is, by the Chowla–Selberg formula for the CM point $\tau = i$,*

$$\eta(i) = \frac{\Gamma(1/4)}{2\pi^{3/4}}. \quad (12)$$

3. *The arithmetic-geometric mean of 1 and $\sqrt{2}$ is the reciprocal of $\sqrt{2}$ times $\eta(i)^2$:*

$$\text{AGM}(1, \sqrt{2}) = \frac{1}{\sqrt{2}\eta(i)^2}. \quad (13)$$

Proof. Statement (1) is a property of the square lattice: the modular parameter $\tau = \omega_2/\omega_1$ is the ratio of the two periods. For the square lattice the periods are equal in magnitude and orthogonal, so $\tau = i$. Statement (2) is the classical Chowla–Selberg formula for the CM point $\tau = i$ [12, 13]. Statement (3) follows from (2) and the classical identity

$$\text{AGM}(1, \sqrt{2}) = \frac{\sqrt{2}\pi^{3/2}}{\Gamma(1/4)^2},$$

which is equivalent to the Legendre relation $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$ and the AGM- K identity $K(k) = \pi/(2 \text{AGM}(1, \sqrt{1-k^2}))$. Substituting (2) into the right-hand side of the claimed formula gives

$$\frac{1}{\sqrt{2}\eta(i)^2} = \frac{2\pi^{3/2}}{\sqrt{2}\Gamma(1/4)^2} = \frac{\sqrt{2}\pi^{3/2}}{\Gamma(1/4)^2} = \text{AGM}(1, \sqrt{2}).$$

Verified at 100-digit precision in Appendix A. □

Corollary 3.5 (Substrate-level form of $I(2)$). *The strip integral $I(2)$ of Theorem 3.1 is expressible as a substrate-level quantity:*

$$I(2) = 1 - \frac{1}{2\sqrt{2}\eta(i)^2}. \quad (14)$$

The transcendental content of $I(2)$ is carried entirely by $\eta(i)$, the Dedekind eta function at the self-dual modular point of the square torus.

Proof. Substitute Theorem 3.4(3) into Theorem 3.1. □

Remark 3.6 (Why the derivation is substrate-level). *Theorem 3.1 derived the AGM identity analytically, from the Beta function $B(-1/4, 1/2)$. Theorem 3.4 derives the same identity from the modular structure of the 8×8 lattice. The two derivations are independent. The substrate-level*

derivation uses only the lattice shape (square), the modular parameter ($\tau = i$), and the Chowla–Selberg formula at the CM point $\tau = i$. None of these is a continuum import. The lattice shape is a property of $\mathbb{Z}_8 \times \mathbb{Z}_8$. The modular parameter $\tau = i$ is a shape invariant, not a size invariant: it is the same for the 8×8 lattice, the 80×80 lattice, and the infinite square lattice. The Chowla–Selberg formula is a theorem about CM elliptic curves, which the square lattice realizes.

Remark 3.7 (Discrete vs. continuous). *A strict reading might object that the Dedekind eta function is a continuum object. The answer is that $\eta(\tau)$ depends only on the modular parameter τ , which is fixed by the lattice’s shape. The 8×8 lattice is square, hence $\tau = i$, hence $\eta(i)$ is well-defined as a modular invariant of the lattice. The discrete lattice’s natural cutoff at $n = 8$ (the side length) truncates the q -series $\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n)$ at $n = 8$, and the truncation error at $q = e^{-2\pi}$ is below 10^{-50} . The discrete and continuous values of $\eta(i)$ are therefore indistinguishable at any physical precision.*

Remark 3.8 (Two independent derivations of $\text{AGM}(1, \sqrt{2})$). *Theorem 3.1 (analytic) and Theorem 3.4 (substrate-level) give the same value of $\text{AGM}(1, \sqrt{2})$. Their agreement is a consistency check on both. The substrate-level derivation (Theorem 3.4) is the HCSM-native one; the analytic derivation (Theorem 3.1) is the classical one.*

4 Crossover-Zero Moments

4.1 The Brillouin-zone structure

The discrete 5-point Laplacian on the $L \times L$ torus has eigenvalues

$$\lambda_L(k_1, k_2) = 4 - 2 \cos(2\pi k_1/L) - 2 \cos(2\pi k_2/L), \quad k_1, k_2 \in \{0, \dots, L-1\}. \quad (15)$$

In the continuum limit $L \rightarrow \infty$, this becomes

$$\lambda(k_1, k_2) = 4 - 2 \cos k_1 - 2 \cos k_2, \quad (k_1, k_2) \in [0, 2\pi]^2, \quad (16)$$

with BZ average

$$\langle f \rangle_{\text{BZ}} = \int_{[0, 2\pi]^2} f(\lambda(k)) \frac{d^2 k}{(2\pi)^2}. \quad (17)$$

4.2 The three crossover points

The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ has three natural crossover points in the BZ: $\lambda = 0$ at the corner $\Gamma = (0, 0)$, $\lambda = 4$ at the self-paired points $X = (\pi, 0)$, $(0, \pi)$, and $\lambda = 8$ at the corner $M = (\pi, \pi)$. An integrand is called *crossover-vanishing* if it vanishes at all three of these λ values.

4.3 The crossover moments

Theorem 4.1 (Crossover-zero moments). *The Brillouin-zone moments of the three crossover-zero integrands on the torus are exactly*

$$\langle (\lambda - 4)^2 \rangle_{\text{BZ}} = 4, \quad (18)$$

$$\langle \lambda(8 - \lambda) \rangle_{\text{BZ}} = 12, \quad (19)$$

$$\langle (\lambda - 4)^2 \lambda(8 - \lambda) \rangle_{\text{BZ}} = 28. \quad (20)$$

Proof. Each moment is a polynomial in λ . Using the standard BZ moments $\langle \lambda \rangle = 4$, $\langle \lambda^2 \rangle = 20$, $\langle \lambda^3 \rangle = 112$, $\langle \lambda^4 \rangle = 656$, the polynomial expansions give the stated values. $\square$

Corollary 4.2 (Mersenne pattern). *The crossover-zero moments satisfy $\langle (\lambda - 4)^2 \rangle_{\text{BZ}} = d(2^1 - 1) = 4$, $\langle \lambda(8 - \lambda) \rangle_{\text{BZ}} = d(2^2 - 1) = 12$, $\langle (\lambda - 4)^2 \lambda(8 - \lambda) \rangle_{\text{BZ}} = d(2^3 - 1) = 28$, with $d = 4$.*

Corollary 4.3 (Mirror ratio). *The ratio of the two lowest crossover-zero moments is*

$$\frac{\langle (\lambda - 4)^2 \rangle_{\text{BZ}}}{\langle \lambda(8 - \lambda) \rangle_{\text{BZ}}} = \frac{4}{12} = \frac{1}{3} = \frac{1}{d - 1}. \quad (21)$$

5 The Discrete Analogue: Negative Result

5.1 The S6 spectral sum

The natural candidate for a discrete analogue of $I(2)$ is the Brillouin-zone average

$$S6(L) = \frac{1}{L^2} \sum_{k_1, k_2=0}^{L-1} \frac{\sqrt{\lambda_L(k_1, k_2)}}{1 + \lambda_L(k_1, k_2)}. \quad (22)$$

5.2 The near-miss

Numerically, $S6(L)$ converges to $S6(\infty) = 0.40083787\dots$, which is 0.023% below $I(2) = 0.40092988\dots$. The residual is 9.2×10^{-5} .

Theorem 5.1 (Negative result on the discrete analogue). *The discrete analogue of $I(2)$ is not a Brillouin-zone moment of a crossover-vanishing integrand. Specifically:*

1. *The S6 spectral sum satisfies $S6(\infty) = I(2) - 9.20 \times 10^{-5}$; the residual is 0.023% of $I(2)$.*
2. *An exhaustive search over ~ 80 framework expressions finds no framework constant whose ratio to the residual is a small integer.*
3. *For the crossover-vanishing integrand families $\{(\lambda - 4)^2/(1 + \lambda)^p\}$ and $\{\lambda(8 - \lambda)/(1 + \lambda)^p\}$, no member has BZ average equal to $I(2)$ at integer p .*
4. *The strip entanglement entropy of the free-fermion Ω^1 Fermi sea on the 8×8 torus scales as a volume law, not the log-law of a 2D CFT.*

Remark 5.2 (Interpretation). *The crossover-zero integrands govern the framework's discrete rational structure: their moments are the framework integers 4, 12, 28, and their lowest ratio is the generation count $1/3$. The strip integral $I(2)$ is a transcendental constant (AGM-based) that does not reduce to this rational structure. This is a genuine structural separation: the substrate's discrete rational skeleton is not the source of the strip integral.*

5.3 What the negative result does and does not rule out

The negative result rules out the free-fermion Ω^1 Fermi sea strip entropy as the discrete analogue, the S6 spectral sum as an exact BZ moment, and any single-moment construction over the crossover-vanishing integrand family.

The negative result does *not* rule out a discrete analogue that lives in a different sector, involves the DNLS-interacting ground state, requires nontrivial state-preparation, or only emerges in a different continuum limit.

Since the present paper was first drafted, one of these possibilities has been realized. The substrate-level derivation of $\text{AGM}(1, \sqrt{2})$ in Section 3.4 proceeds through the modular structure of the self-dual square torus, not through a BZ moment. The derivation uses the Dedekind eta function at $\tau = i$ and the Chowla–Selberg formula. The transcendental content of $I(2)$ is therefore derived from the substrate without invoking a free Ω^1 BZ moment. The negative result of Theorem 5.1 remains valid; what changes is the interpretation: the BZ-moment route was not the right route, but the modular route succeeds.

6 The Vacuum Stability Theorem

Theorem 6.1 (Vacuum stability). *The equilibrium dimension $D = 4$ is a strict local maximum of the vacuum entropy. The second derivative at the equilibrium is*

$$S''(0)/S_0 = -2\gamma = -1.405678935980144918585156393018\dots < 0. \quad (23)$$

The equilibrium is unique: no other integer D in the range $1 \leq D \leq 8$ gives a strict local maximum with the same numerical value of $S''(0)/S_0$.

Proof. The entropy well (2) has second derivative -2γ at $\epsilon = 0$. The value of γ is established in Theorem 3.3. Numerical verification is in Appendix A. $\square$

Corollary 6.2 (Downstream predictions from γ). *The EFT vacuum equation of state is*

$$w = -1 - \frac{(1 - \gamma)^2}{2\pi} = -1.0140540788576648688\dots, \quad (24)$$

and the bulk viscosity is

$$\frac{\zeta H}{\rho} = \frac{(1 - \gamma)^2}{6\pi} = 0.004684692952554956\dots \quad (25)$$

Both are consistent with DESI DR2 at $\Delta\chi^2 = 0.22$.

7 Status: Derived, Disclosed Identification, and Open

7.1 Derived (substrate-level)

1. The 8×8 torus is square; $\tau = i$.
2. $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$ (Chowla–Selberg).
3. $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$.
4. $I(2) = 1 - 1/(2\sqrt{2}\eta(i)^2)$.
5. The crossover-zero moments 4, 12, 28.
6. The Mersenne pattern $d(2^n - 1)$ for $n = 1, 2, 3$.
7. The mirror ratio $1/3 = 1/(d - 1)$.
8. $S''(0)/S_0 = -2\gamma < 0$.
9. The negative result on the discrete BZ analogue.

7.2 Derived (analytic)

10. The AGM identity $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$.
11. The closed form of $c_2 = 2I''(2)$.
12. The closed form of $\gamma = c_2/I(2)$.

7.3 Disclosed identification

13. The identification $\gamma = c_2/I(2)$.

7.4 Open

14. The physical mechanism by which c_2 enters at second order.
15. The finite- L corrections to the crossover-zero moments.
16. The connection to the sigma-line structure of HCSM-24.

8 Falsifiable Predictions

1. **Equation of state.** $w = -1 - (1 - \gamma)^2/(2\pi) = -1.014054\dots$, consistent with DESI DR2 at $\Delta\chi^2 = 0.22$ and testable at $> 3\sigma$ by DESI DR3 + Euclid.
2. **Bulk viscosity.** $\zeta H/\rho = 0.00468469\dots$, entering the framework's EFT predictions for the dark-energy density perturbations.
3. **Strict local maximum.** Any deviation $\epsilon = D - 4$ is unstable to decay toward $\epsilon = 0$. A detection of a stable equilibrium at $D \neq 4$ would falsify the framework.
4. **Negative prediction on the discrete analogue.** No BZ moment of a crossover-vanishing integrand on the discrete torus equals $I(2)$ exactly.
5. **Modular structure.** The framework predicts that the 8×8 torus is self-dual ($\tau = i$), with $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$. A measurement that determines the effective modular parameter of the substrate to be different from $\tau = i$ would falsify Theorem 3.4.

9 Conclusion

We have established the vacuum stability sector of the Hodge Complex Standard Model. The stability coefficient

$$\gamma = c_2/I(2) = 0.70283946799007245929257819650854315216110316215188\dots$$

controls the width of the entropy well around $D = 4$, and the strict-local-maximum property $S''(0)/S_0 = -2\gamma < 0$ is the vacuum stability theorem.

The AGM identity $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ reduces the strip integral to a specific arithmetic constant. The substrate-level derivation of $\text{AGM}(1, \sqrt{2})$ from the self-dual square torus shows that this constant is a modular invariant of the 8×8 lattice. The closed form of $c_2 = 2I''(2)$ in π , $\Gamma(1/4)$, and Catalan's G completes the derivation of γ as a specific transcendental constant. The

crossover-zero moments 4, 12, 28 on the discrete torus are exact framework integers whose lowest ratio $1/3 = 1/(d-1)$ is the generation count.

A negative result is also established. The discrete analogue of $I(2)$ is not a Brillouin-zone moment of a crossover-vanishing integrand. The closest candidate, the S6 spectral sum, is 0.023% below $I(2)$, and the residual is not a framework constant.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

References

- [1] S. Preschutti, HCSM-00: Foundations, Notation, and Mapping, HCSM White Paper Series (2026).
- [2] S. Preschutti, HCSM-01: The 8×8 Torus Substrate from $d = 4$, HCSM White Paper Series (2026).
- [3] S. Preschutti, HCSM-02: The Hodge Complex on Λ , HCSM White Paper Series (2026).
- [4] S. Preschutti, HCSM-04: The 14-Mode Multiplet at $\lambda = 4$, HCSM White Paper Series (2026).
- [5] S. Preschutti, HCSM-20: The Motion Theorem, HCSM White Paper Series (2026).
- [6] S. Preschutti, HCSM-21: The Time Theorem, HCSM White Paper Series (2026).
- [7] S. Preschutti, HCSM-24: The Charge Theorem, HCSM White Paper Series (2026).
- [8] S. Preschutti, HCSM-51: The kL Crossover, HCSM White Paper Series (2026).
- [9] S. Preschutti, HCSM-52: EFT Coefficient in ρ_{DE} , HCSM White Paper Series (2026).
- [10] S. Preschutti, HCSM-56: Vacuum EFT, HCSM White Paper Series (2026).
- [11] J. M. Borwein and P. B. Borwein, *Pi and the AGM*, Wiley (1987).
- [12] S. Chowla and A. Selberg, On Epstein's zeta-function, *J. Reine Angew. Math.* **227**, 86 (1967).
- [13] M. Lerch, Sur quelques formules relatives au nombre des classes de formes quadratiques binaires, *Acta Math.* **16**, 275 (1892).
- [14] I. Peschel, Calculation of reduced density matrices from correlation functions, *J. Phys. A* **36**, L205 (2003).
- [15] J. Dodziuk, Finite-difference approach to the Hodge theory of harmonic forms, *American Journal of Mathematics* **98**, 79 (1976).
- [16] DESI Collaboration, DESI DR2 Results, arXiv:2503.14738 (2025).

A Reproducible verification

The verification script `12-VERIFICATION/HCSM-54_verification.py` verifies:

1. The AGM identity $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ at 100-digit precision.
2. The closed form of $c_2 = 2I''(2)$.
3. The closed form of $\gamma = c_2/I(2)$.
4. The substrate-level AGM identity $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$ and the Chowla–Selberg value $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$.
5. The substrate-level form $I(2) = 1 - 1/(2\sqrt{2}\eta(i)^2)$.
6. The crossover-zero moments 4, 12, 28.
7. The mirror ratio $1/3$.
8. The S6 spectral sum convergence and the 0.023% residual.
9. The exhaustive gap search over ~ 80 framework expressions.
10. The strip entanglement entropy of the free Ω^1 Fermi sea and the volume-law scaling.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

B Web Crawler Metadata

B.1 Dublin Core

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DC.description: We establish the vacuum stability sector of the Hodge Complex Standard Model. The stability coefficient is $\gamma = c_2/I(2) = 0.70283946799007245929257819650854315216110316215188\dots$. The strip integral at $\alpha = 2$ has the exact closed form $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$, verified at 100 digits. The AGM constant is a substrate-level invariant of the self-dual square torus: $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$ with

$\eta(i) = \Gamma(1/4)/(2 \pi^{3/4})$. The crossover-zero moments on the discrete torus are exactly 4, 12, 28, and the mirror ratio $1/3$ is the generation count.

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     $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ . The AGM constant is a substrate-level
    invariant of the self-dual square torus. The crossover-zero moments
    on the discrete torus are exactly 4, 12, 28, and the mirror ratio
     $1/3$  is the generation count.}
}
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